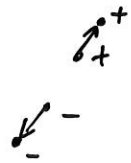
courtesy of J. Skowronek

Drude model of metals

Tues 25 Feb

$\vec{P}$ - polarization density

$$\vec{J} = \frac{\partial \vec{P}}{\partial t} + \nabla \times \vec{M}$$

1- 

2- Metals

1- Drude model for a metal. $\vec{E}_0(t) \rightarrow \vec{P}(t)$ - ?

2- $\vec{E}_0(t) \longrightarrow \vec{E}_0(\omega)$ solve for $\vec{P}(\omega)$

3- $\mathcal{F}^{-1} \longrightarrow \vec{P}(t)$

Drude model ignores Coulomb interactions between e^-'s

$$m\vec{a} = q\vec{E} - \frac{m}{\tau}\vec{v}$$

$$m\dot{\vec{v}} + \frac{m}{\tau}\vec{v} = q\vec{E}_0 e^{-i\omega t}$$

$$\vec{v}(t) = \vec{v}_0 e^{-i\omega t} \longrightarrow \dot{\vec{v}} = -i\omega \vec{v}_0 e^{-i\omega t}$$

$$\left(-i\omega m + \frac{m}{\tau}\right)\vec{v}_0 e^{-i\omega t} = q\vec{E}_0 e^{-i\omega t}$$

$$\vec{v}_0 = \frac{q\vec{E}_0}{\left(\frac{m}{\tau} - i\omega m\right)}$$

Homogeneous soln decays fast, so we only worry about steady-state soln.

$$\vec{v} \xrightarrow{?} \vec{J} \longrightarrow \vec{J} = \rho\vec{v} \qquad \rho = nq$$

$$-\oint_{C_1} \frac{d\omega}{(\omega+i\xi)(\omega+i/\tau)} e^{-i\omega(t-t')} = 2\pi i \sum_j \mathrm{Res}(j) = 2\pi i\left[\mathrm{Res}(i\xi) + \mathrm{Res}(i/\tau)\right]$$

$$f(\omega) = \frac{g(\omega)}{(\omega - p)}$$

$$\mathrm{Res}\{f(p)\} = g(p)$$

$$\mathrm{Res}(-i\xi) = \lim_{\omega \to -i\xi} \frac{e^{-i\omega(t-t')}}{(\omega + i/\tau)}$$

$$= \frac{e^{-\xi(t-t')}}{i\left(\frac{1}{\tau} - \xi\right)} \to \lim_{\xi \to 0} \to -i\tau$$

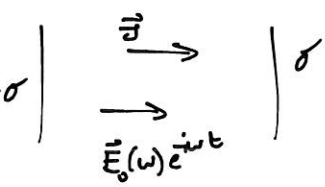
$$\mathrm{Res}(-i/\tau) = \frac{e^{-(t-t')/\tau}}{i\left(\xi - \frac{1}{\tau}\right)} \longrightarrow \lim_{\xi \to 0} \longrightarrow i\tau e^{-(t-t')/\tau}$$

$$\boxed{= \frac{2\pi n q^2}{2\pi m} \int_{-\infty}^{t} \vec{E}_0(t')dt'\left(\tau\left(1 - e^{-(t-t')/\tau}\right)\right)}$$

$P_\omega = \chi_\omega \vec{E}_\omega \longrightarrow$ ω-dispersion $\longrightarrow \int$ in time

$\chi_\omega \equiv$ susceptibility

But at DC $\omega \to 0$ $P_\omega \to \infty$, so it's necessary to maintain finite bounds such that $\vec{J} = \Sigma q \vec{r} \not\to \infty$



$$JA = \frac{\sigma A}{t} \longrightarrow \boxed{J = \frac{d\sigma}{dt}}$$

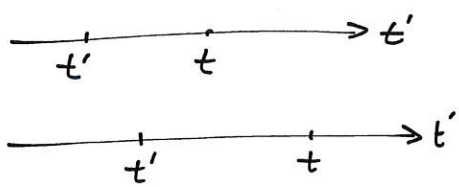
Thurs 27 Feb

More details regarding Fourier shortcuts from last time.

Response of stimulus in UE is instantaneous, e.g. $\vec{P}(t) = \varepsilon_0 \chi_e \vec{E}$

In reality,

$$\vec{P}(t) = \varepsilon_0 \int_{-\infty}^{\infty} dt' \, \chi(t,t') \vec{E}(t') \quad \text{(Kernel)}$$



As long as system is time-invariant,

$$\chi(t,t') = \chi(t-t')$$

$$\vec{P}(\omega) = \int_{-\infty}^{\infty} dt \, \vec{P}(t) e^{i\omega t} = \varepsilon_0 \int_{-\infty}^{\infty} dt' \int_{-\infty}^{\infty} dt \, \chi(t-t') \vec{E}(t') e^{i\omega(t-t')} e^{i\omega t'}$$

$$= \varepsilon_0 \chi(\omega) \vec{E}(\omega)$$

$$\vec{P}(t) = \vec{P}(\omega) e^{-i\omega t} = \varepsilon_0 \chi(\omega) e^{-i\omega t} \vec{E}(\omega)$$

Cont'd from before ...

$$\boxed{\vec{J}(\omega) = \frac{ie^2 n}{m(\omega + i/\tau)} \vec{E}_{tot}(\omega)} \qquad J(t) = \frac{\partial P(t)}{\partial t} \longrightarrow J(\omega) = -i\omega P(\omega)$$

Current response to local (total) electric field

$$\vec{P}(\omega) = -\varepsilon_0 \frac{\omega_0^2}{\omega(\omega + i/\tau)} \vec{E}_{Tot}(\omega)$$

$$\vec{P}(\omega) = -\varepsilon_0 \frac{\omega_0^2}{\omega(\omega + i/\tau) - \omega_0^2} \vec{E}_0(\omega)$$

$$\vec{E}_{Tot}(\omega) = \frac{\omega(\omega + i/\tau)}{\omega(\omega + i/\tau) - \omega_0^2} \vec{E}_0(\omega)$$

$$\vec{P}(t) = ? = \frac{-\varepsilon_0 \omega_0^2}{2\pi} \int_{-\infty}^{\infty} \frac{d\omega\, \vec{E}_0(\omega) e^{-i\omega t}}{\omega(\omega + i/\tau) - \omega_0^2}$$

$$= \frac{-\varepsilon_0 \omega_0^2}{2\pi} \int_{-\infty}^{\infty} dt'\, \vec{E}_0(t') \int_{-\infty}^{\infty} \frac{d\omega\, e^{-i\omega(t-t')}}{\omega(\omega + i/\tau) - \omega_0^2}$$

$$\omega(\omega + i/\tau) - \omega_0^2 = 0 \longrightarrow \omega^2 + \omega\, i/\tau - \omega_0^2 = 0$$

$$\text{poles} = -\frac{i}{2\tau} \pm \omega_0 \qquad \omega_0 \gg \frac{1}{\tau}$$

<u>$t > t'$</u>

$$\int_{-\infty}^{\infty} \frac{d\omega\, e^{-i\omega(t-t')}}{(\omega + i/\tau + \omega_0)(\omega + i/\tau - \omega_0)} = 2\pi i \left[\mathrm{Res}\left(-\frac{i}{2\tau} + \omega_0\right) + \mathrm{Res}\left(-\frac{i}{2\tau} - \omega_0\right) \right]$$

$$= 2\pi i \left[\frac{e^{-i\left(\frac{-i}{2\tau} + \omega_0\right)(t-t')}}{2\omega_0} - \frac{e^{-i\left(\frac{-i}{2\tau} - \omega_0\right)(t-t')}}{2\omega_0} \right]$$